Yes, the N=96 model computes DD's deducibility predicate — in its last layer

6 checkpoints x 3 probe seeds · jobs 50979059 + 51009233 · commit 309ad1a · linear probes on the residual stream

THE PREDICATE IS THERE: AUROC 0.731-0.772 against a label-only free score of 0.673.

frontier_mask is which nodes DD can deduce this round. A reader that knows only which nodes are already coloured scores 0.6734 on it for free (spread 0.0050 over 3 instance pools); a probe handed n_known_colors -- the quantity DD actually thresholds -- reaches 0.808. All six cells land inside that bracket, closing 43%-73% of it on AUROC and 57%-83% on average precision. This is beyond what bookkeeping explains -- and well short of the information DD uses.

IT IS BUILT LAST, and the control says that is about DD and not about the probe.

frontier_mask peaks at layer 8, 12 and 16 for L=8, 12 and 16 -- fraction 1.000, spread 0.000, identical across three probe seeds AND across training seeds. known_mask, pure bookkeeping, is already at 0.768 from the EMBEDDING and peaks at layers 4/8/2 with no depth pattern at all. The deduction predicate and the bookkeeping are different signals living in different places.

BUT MORE DEPTH BUYS NO MORE OF IT.

Peak lift over the embedding is 0.143 / 0.119 / 0.131 at L=8 / 12 / 16 -- spread 0.0237, under the 0.0411 one-training-seed yardstick. Eight extra layers move where the predicate tops out and not how much of it there is. Consistent with the scan, where the same depth range was also unreadable against the seed.

AND IT DOES NOT PREDICT SUCCESS: Spearman -0.086 vs rho_50, while next-node choice gives -0.943.

The worst model in the grid (t5, rho_50 0.829) has the SECOND-HIGHEST frontier AUROC. What separates good cells from bad is next_node -- which of the deducible nodes the model commits to emitting -- where t5 collapses to 0.603 against 0.870. Every cell works out roughly what is deducible; they differ in acting on it.

Probe eval: 400 train / 160 test instances, <=20 reasoning positions each, 600 probe epochs, three probe-fit seeds. AUROC, not balanced accuracy -- page 4 is about why that choice reverses the verdict. rho_50 from the same 14-point eval as the scan deck.

What the model actually sees

Every token is one categorical id into a 106-symbol vocabulary (N=96, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-105	`N0 N1 ... N95`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`
`SOLVED`	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 790 TOKENS

HINTS	`BEGIN_HINTS N1 COLOR_0 N3 COLOR_1 N5 COLOR_0 N10 COLOR_2 N14 COLOR_0 N18 COLOR_0 ... (71 tokens)`
GRAPH	`BEGIN_GRAPH N0 N34 N73 N78 N19 N80 N1 N37 N56 N31 N36 N61 ... (577 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N6 COLOR_3 N7 COLOR_1 N9 COLOR_1 N19 COLOR_1 N22 COLOR_1 N28 ... (142 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 18 rounds, 61 deductions,
122 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000016, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

35 hints of 96 nodes, then 18 rounds of simultaneous deduction. Hints: N1=COLOR_0 N3=COLOR_1 N5=COLOR_0 N10=COLOR_2 N14=COLOR_0 N18=COLOR_0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N6 COLOR_3 N7 COLOR_1 N9 COLOR_1 N19 COLOR_1 ...
  13 deduced
BEGIN_ROUND        N0 COLOR_0 N16 COLOR_0 N42 COLOR_3 N44 COLOR_1 ...
  6 deduced
BEGIN_ROUND        N4 COLOR_1 N11 COLOR_3 N55 COLOR_2 N69 COLOR_3 ...
  5 deduced
BEGIN_ROUND        N2 COLOR_2 N15 COLOR_0 N58 COLOR_0
  3 deduced
... 14 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N6, neighbours N76 N12 N80 N89 N87

their colours known before this round:

```
   [0, 1, 2]  (3 = q-1 of the 4)

   (3 known neighbours, 3 DISTINCT colours)

   still unknown: N76 N12
```

only colour left: 3

```
so the trace emits  N6 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

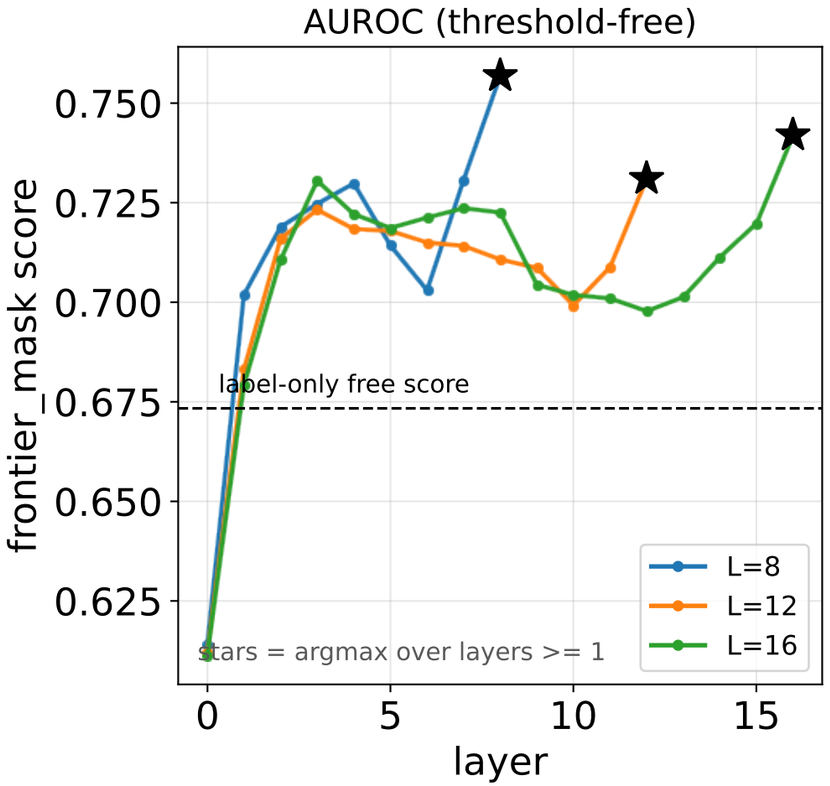
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

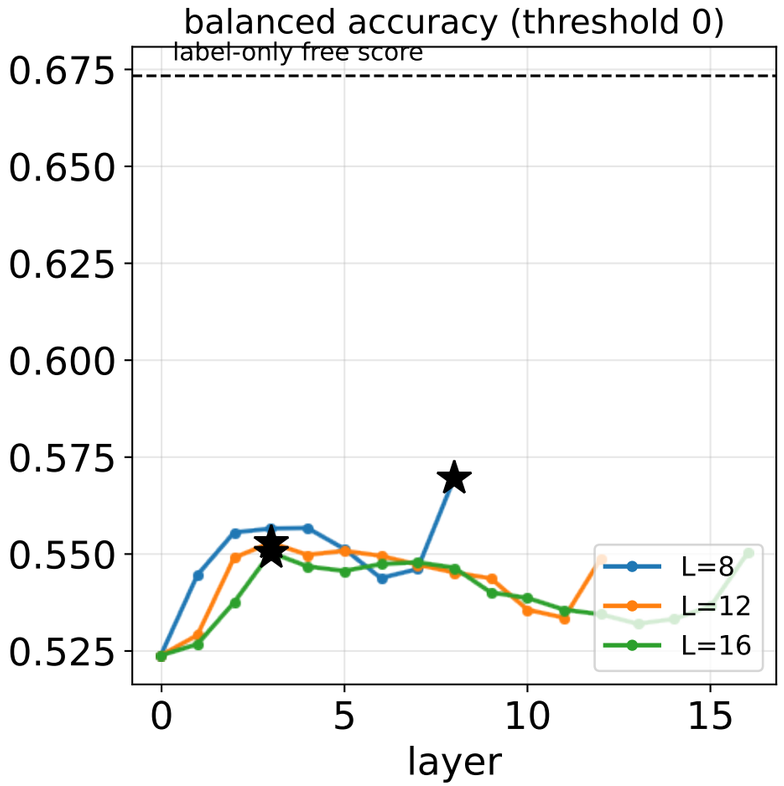
N=96, degree 5, 4 colours, --shuffle-adj --fall: the task all six probed checkpoints were trained on. The probe reads the residual stream at the reasoning positions -- one per node/colour emission, capped at 20 per instance.

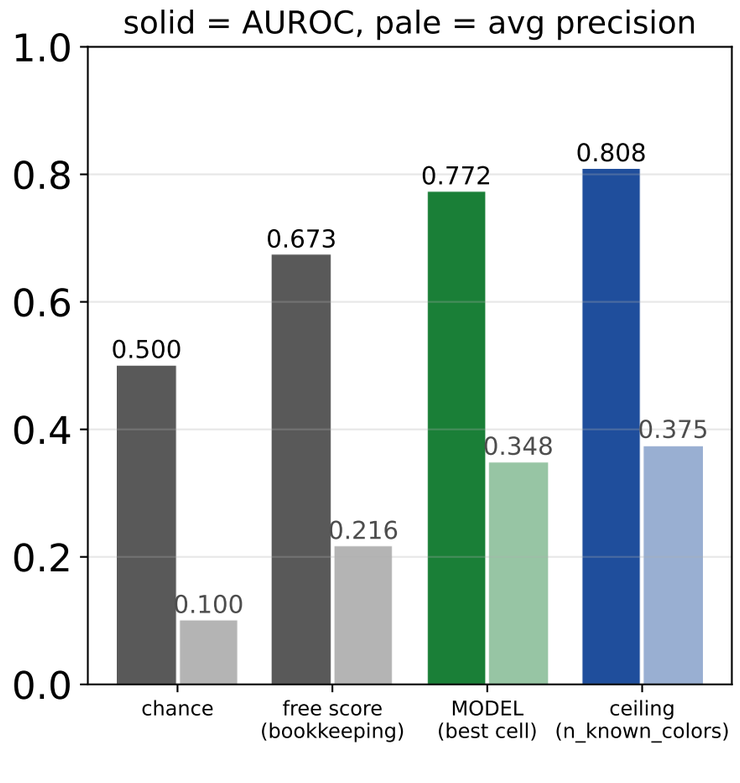
T=18 rounds, the 90th percentile of 40 draws from this config (range 1-23) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Read on the wrong column, this run is a null

Same CSVs, same rows, two columns. frontier_mask is positive on 10.0% of bits.







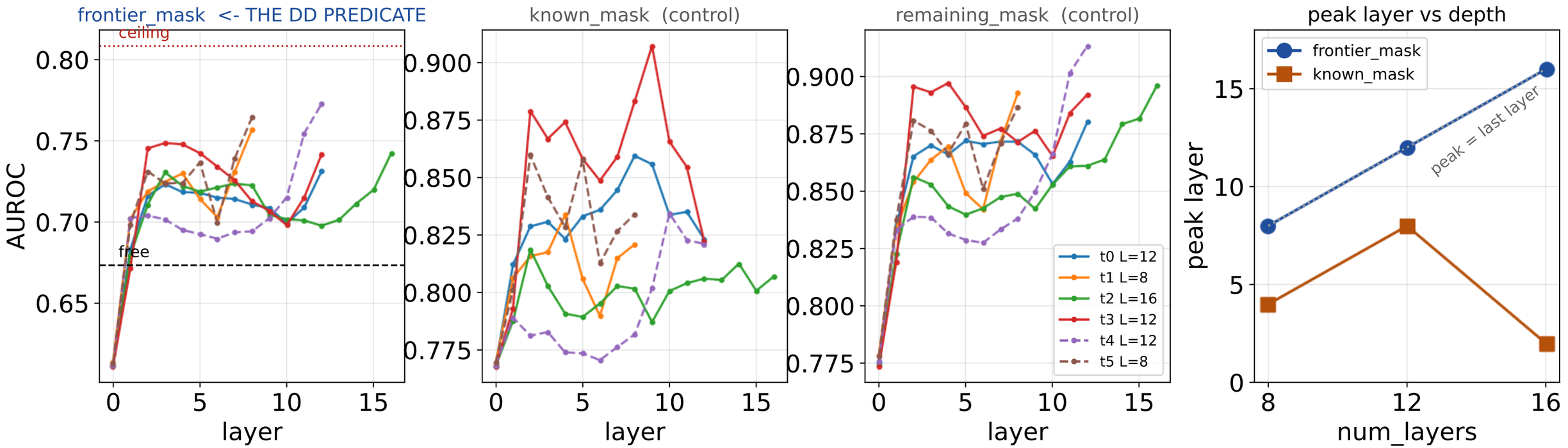
A BCE probe on a 10%-positive target under-predicts positives, so its balanced accuracy at threshold 0 is a FLOOR that moves with calibration, not with knowledge. The tell was already in the baseline: the TRIVIAL rule 'not yet coloured' scores 0.6734 there, above every trained probe. That cannot be a statement about knowing.

Same trap at the other end: DD's own rule, computed not fitted, scores only 0.765 AUROC -- a 0/1 rule gets no credit for ranking within its two classes -- but 0.490 average precision, the highest of anything here. The graded ceiling is color_aug, same information, allowed to rank.

Read on balanced accuracy, every cell sits below the free score and the peak layer wanders 8.67 layers across L=8/12/16 -- while one training seed alone moves it 9.00 at fixed L=12. That is a flat curve's argmax. The baseline script's own docstring had already said to quote the threshold-free columns.

Where DD gets done: bookkeeping early, the predicate last

Layer 0 is the token+position embedding, so it is the tokenization control: whatever it scores needs no computation.



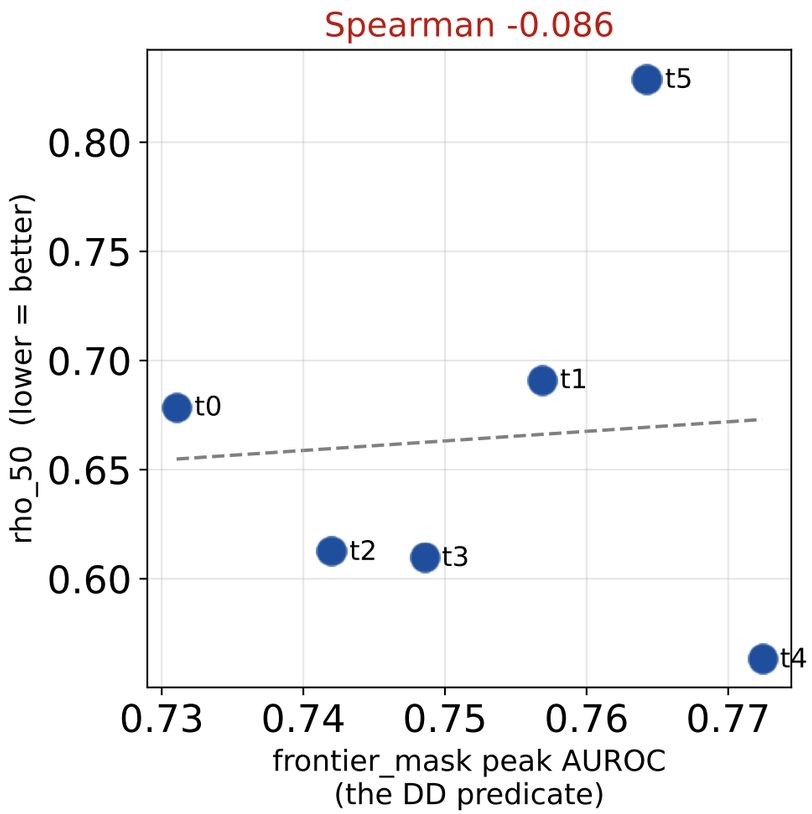
P1 as pre-registered FAILS (the peak is not at a fixed absolute layer, it moves 8) — the informative alternative holds in its extreme form:
a fixed FRACTION of depth, and the fraction is 1.000 with spread 0.000. P3, the control clause, passes: known_mask shows no depth pattern (spread 0.542).

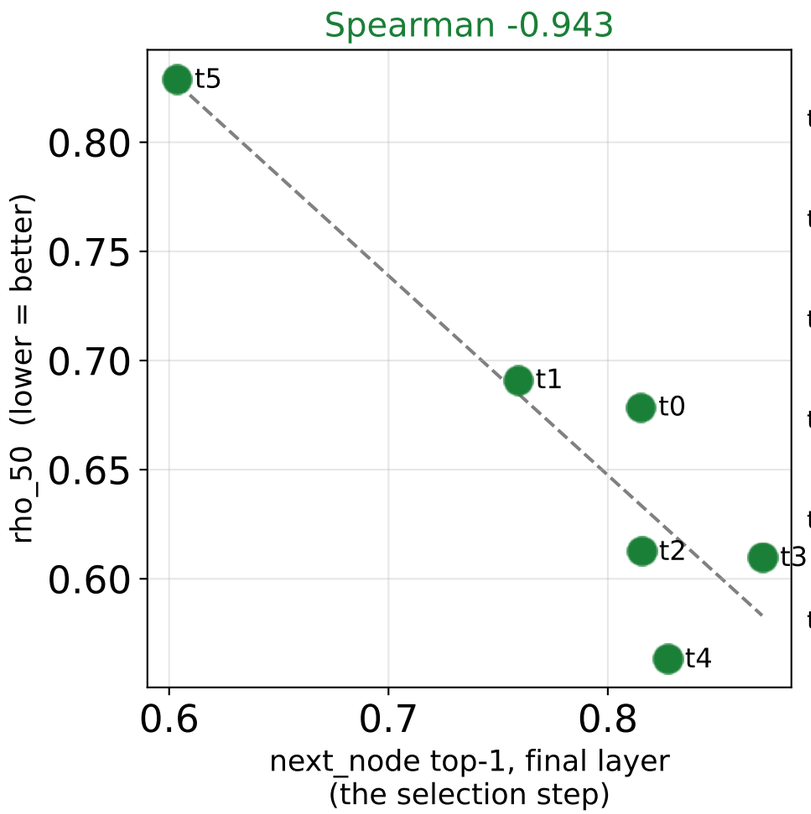
Noise floors both claims were read against — P2(a) three probe-fit seeds: frontier peak layer spread 0.00, lift 0.0090.
P2(b) training-seed pairs: peak layer spread 0.00, lift 0.0411. The LAYER is solid; the AMOUNT is not, which is why no depth claim is made about the amount.

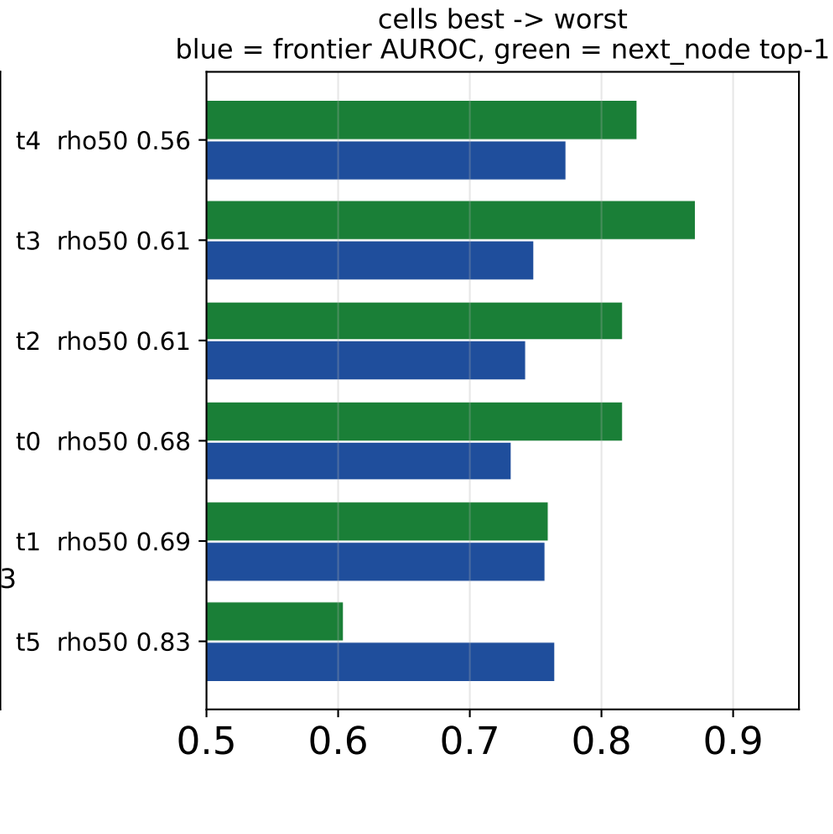
known_mask reaching 0.768 at layer 0 is not a result about the model: the token sequence states which nodes have been coloured, so the embedding carries it. That is exactly why it is the control -- any layer structure it showed would be the probe's, not DD's.

Knowing what is deducible is not what separates the models

Six cells, each with rho_50 from the 14-point x 400-instance eval (lower = solves with fewer hints = better).







t5 is the worst model in the grid (rho_50 0.829, val_nll 1.814) and has the SECOND-HIGHEST frontier AUROC (0.764).
Its next_node collapses to 0.603 against 0.870 for the best cell. The predicate channel is flat across good and bad models; the selection channel is not.

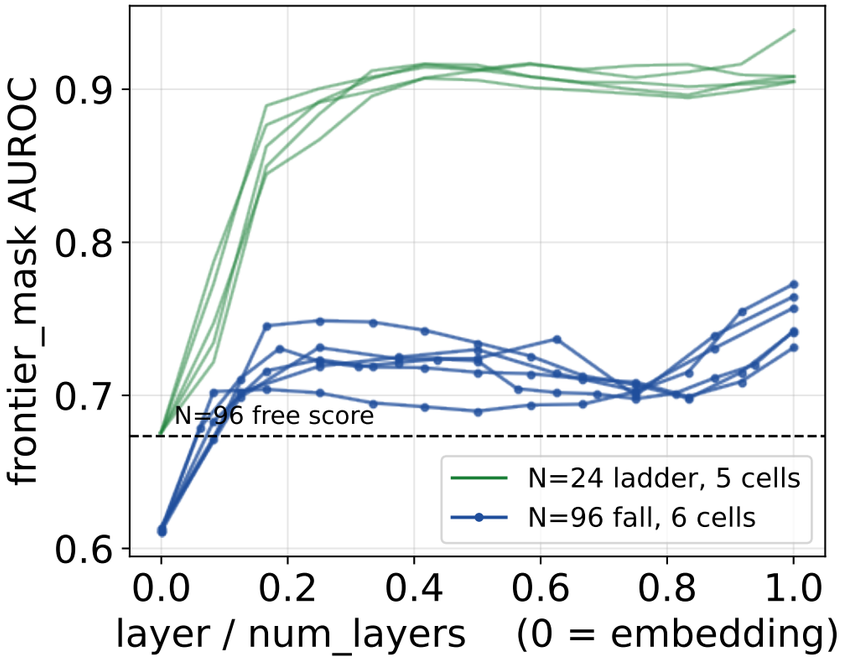
Two caveats kept in view: n=6, and next_node carries a 0.1555 training-seed spread at L=8, so -0.943 is an observation over six points, not an estimate. next_node also sits closer to the logits than frontier_mask does, so part of its edge may be proximity rather than mechanism. The clean test is an INTERVENTION.

This is the finding that redirects the next experiment. The standing question was whether the network runs DD; the probe says it largely works out WHAT is deducible and that this is not where the N=96 models differ. The difference is in which deducible node they then choose to emit.

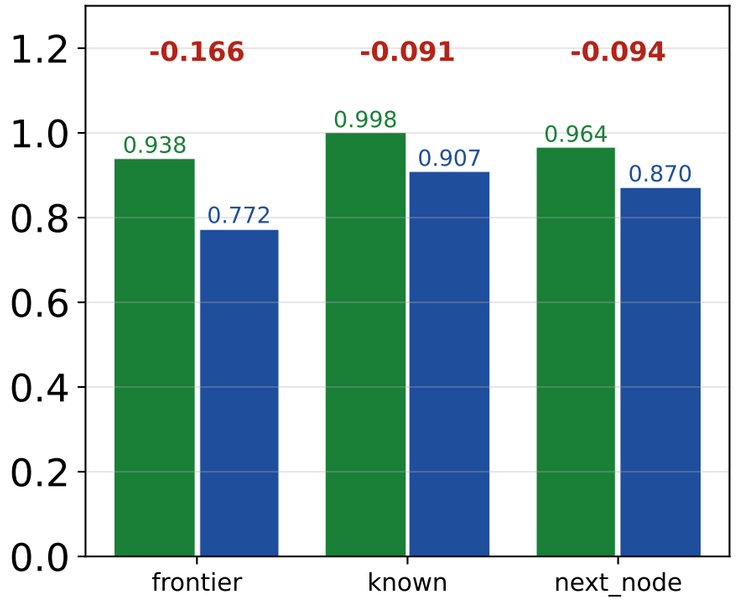
Below the previous best on this eval — and normalising does not rescue it

N=24 competence ladder (2026-10-21), same sudoku_probes code path, same targets, same three probe seeds, L=12. Each N now has a label-only bracket of its own.

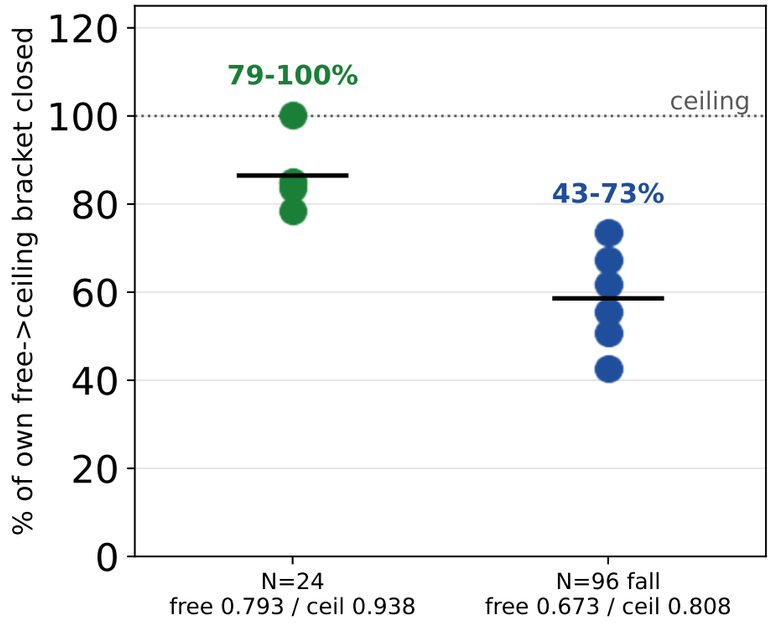
N=24 peaks mid-stack and plateaus;
N=96 rises into the last layer



raw AUROC: N=24 (green) vs N=96 (blue)



normalised by each N's own bracket:
the decline is not just a harder task



N=24 is an easier TASK as well as an easier ensemble — its label-only free score is 0.793 against 0.673 here. Normalising by each bracket, N=24 saturates its ceiling and N=96 does not; and the shortfall is not confined to the predicate — known_mask, pure bookkeeping, near-perfect at N=24, falls to 0.91 at N=96 (no bracket for that one).

Tempting reading — 'N=96 is depth-limited, the computation does not finish inside the stack' — is CONTRADICTED by this very run:
L=16 decodes the predicate no better than L=8 (0.742 vs 0.757, inside the seed yardstick). The last-layer peak is assembly, not an unfinished computation.

NEXT, in order: (1) the causal test — ablate/patch the frontier direction at the final layer and measure rho_50 on the same eval; a probe shows information is readable, not used. (2) Aim it at the SELECTION step, which tracks rho_50 at -0.943, rather than at the predicate, which does not track it at all.
(3) Explain the N=24 -> N=96 bookkeeping collapse, which is upstream of everything here.

Comparability: N=24 is a different ensemble (no fall; frontier positive rate 0.148 against 0.101 here), so even the normalised panel compares two tasks rather than two models on one task, and the AP columns are not comparable across N at all. Both brackets: 3 instance pools, 600 probe epochs, 51009233.